

1. What is a trap in OS?

A trap is a system call that provides OS level service

A trap is a service that user can capture output from a process

A trap is a system call that captures users in OS

A trap is a way to capture bugs

There is no trap in OS

2. What would happen if there is no user level and privileged level separation?

Process cannot be kicked of processor

OS cannot access disks or external devices

Process may be able to access other process data

OS can prevent process from directly access devices

Process cannot create child process

3. Which of the following statements is true?

Processes seem to occupy a CPU provided by modern OS

an instruction in a program directly on disk

A process is a program loaded into memory

All of the statements are true

None of the statements is true

4. What is not the component of a process?

Program

Data state

Process status

Control state

All of the choices are process components

5. What is the issue to consider to protect one application from others?

Performance

Isolation

Reliability

Mobility

All of the issues is for protection

6. What is the low-level hardware that is needed in non-preemptive system to switch process?

MMU

Timer interrupt

Cache

TLB

None of them

7. Which part of the process is in kernel space?

Text segment

Data segment

Stack segment

Process descriptor

All of them are not in kernel

8. Which part of memory of a process grows in an opposite direction of other?

Text

Heap

Stack

Program

They are all growing in the same direction

9. When would a process in ready state be moved to a blocked state?

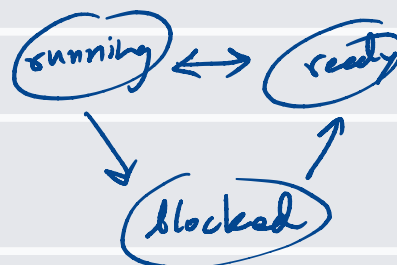
When it read a file

When it read from memory

When timer interrupt occurs

Cannot happen

Both when read from file and memory



10. 1. When would a process enter a ready state?

Timer interrupt occurs

I/O transfer is completed

A process is created

All of the events

None of the events

11. What does a process in blocked state wait for?

I/O transfer to complete

A timer interrupt to occur

A signal from a child process

OS to remove the process from system

OS to create the process and move to ready

12. What does it mean when process is a zombie?

Process is running on processor and cannot be kicked off

Process is waiting for I/O indefinitely

Process is ready but has never be select to run

a process could be placed in a final state where it has exited but has not yet been cleaned up (in UNIX-based systems, this is called the **zombie state**)

Process is terminated but cannot be removed from the system

There is no such thing as zombie in OS

13. What is the data structure that keeps process information?

PCB

TLB

Stack

MMU

Queue

14. In the lab, when a parent waits for the child before printing, what will always happen?

Child will always print after parent

A zombie will always happen

Child will always start from beginning of main function

A segmentation fault will always occur

Parent will always print after child

15. Which of the following scheduling algorithms execute the jobs in the arrival order?

FIFO

SJF

RR

Both FIFO and RR

Both FIFO and SJF

16. Which event can cause a process to release from a CPU?

Timer interrupt

Calling I/O transfer

Calling wait

Calling yield

All of the events

17. What is not true about MLFQ?

It has a number of distinct queues of different priority levels.

Process enters with lowest level of priority

A job that is ready to run is on a single queue

RR is used among jobs in the same queue

All statements are true

18. What is not true about memory?

In multitasking systems, we need physical memory that is larger than virtual memory

Physical Address is provided by the H/W

Logical Address (Virtual Address) is provided by the OS

In multitasking systems, we need process isolation

None of the statements is not true

19. Which of the storage class is the fastest?

Auto

Register

Extern

Static

They are executed at the same speed by CPU

20. Which statement is true?

Single partition allocation is more efficient than multiple partition allocation in terms of memory usage

Flexible-size partition does ~~not~~ suffer external fragmentation

Fixed-size partition suffers from internal fragmentation

External

To solve ~~internal~~ fragmentation, OS needs to do memory coalescing and compaction

None of them is true

21. What does process selection algorithm do?

Select program to load into memory

Select process to run

Select process to remove from memory

Select memory to assign to a process

Select process to put to memory

22. What does page replacement algorithm do?

Select a page to load into memory

Select a page to run

Select a page to remove from memory

Select memory to assign to a page

Select a page to replace a process

23. What is not used in address translation?

MMU

Page table

TLB

Segment table

All of them are used in address translation

24. What is the disadvantage of segmentation with the contiguous memory allocation?

Harder to translate from VA to PA

Cannot protect memory from other process

Suffer from external fragmentation

Use too much memory to keep map of the address translation

There is no disadvantage

25. Which statement is true about virtual memory in windows and Linux?

Both system uses the same demand paging

In Linux, user can request a page using realloc()

Windows system only transfer one page at a page fault

Linux can use swap partition and swap file

Windows only uses swap partition

26. What does demand paging mean?

Page will be loaded when it is accesses only

Page is always in memory for on demand access

Page is pre-loaded on user demand

Page is pre-loaded by probability of the on demand usage

There is no such thing as on demand paging in OS

27. What does not happen when a page fault occurs?

Save process context

Free a frame if there is no free frame

Locate and read page from disk

Update page table

All events happen

28. 1. Which statement is not true?

Windows 10 is using segmentation

Linux is using cluster demand paging

In segmentation we only need one pair of base and bound

In paging, page and frame can be different in size

All of them are false

29. What is the data stored in the segment table?

Base

Bound

Direction

Valid

All choices are a part of segment table

30. What is the effect of a larger page size?

More problem with external fragmentation

Smaller TLB reaches

Smaller page table

More process can be created

There is no effect from the size of page

31. Which situation could lead to zombies in the system?

Parent exists before child returns value

Parent creates more than one child

Child also creates a child process

Parent waits for child

Child executes other program

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32. Which process scheduling has the highest throughput?

FIFO

STCF

RR

MLPQ

a) All of the schedulers have the same throughput

33. Which scheduler is the fairest?

FIFO

STCF

RR

MLPQ

All of the schedulers have the same fairness

34. Which scheduler does not need timer interrupt?

FIFO

STCF

RR

MLPQ

All of the schedulers have the same fairness

35. Where does virtual memory save the image of memory?

On virtual storage

On a virtual cloud

On virtual private network

On kernel of the OS

On secondary storage

36. What is the response time?

Time that process starts running until finish

Time that process enters system until finish

Time that process enters system until running

Time that process runs until requests for I/O

Time that process runs at each cycle

37. What is the turnaround time?

Time that process starts running until finish

Time that process enters system until finish

Time that process enters system until running

Time that process runs until requests for I/O

Time that process runs at each cycle

38. What is the problem of flexible partition size?

Limit degree of multitasking

Internal fragmentation

Address translation

Need compacting

There is no problem

39. What is the problem with paging when address space is larger?

Need more memory to store page table

Internal fragmentation

External fragmentation

TLB reach is smaller

There is no problem caused by larger address space

40. Why multi-level page table could save some memory?

Table size is reduced

Address space is reduced

Page size is increased

All of the second level pages are empty

There is no need to store all the empty second level pages

41. Which state could a process be when it is newly created in OS?

ready

running

blocked

terminated

empty

42. When a process is using a processor, which state could it be?

ready

running

blocked

terminated

empty

43. What is a process in a blocked state waiting for?

User input

Input data from disk

Keyboard input

Context switching

Could be waiting for any of the given answers

44. What is a disadvantage of shortest job first (SJF) scheduling?

Suffer convoy effect

Highest average waiting time

Highest average response time

Highest average turn around time

Starvation

45. What is a disadvantage of shortest job first (SJF) scheduling?

Suffer convoy effect

Highest average waiting time

Highest average response time

Highest average turn around time

Starvation

46. What is the advantage of round robin (RR) process scheduling?

Lowest average turn around time

Lowest average waiting time

starvation could occur

Lowest average response time

highest throughput

47. What is not true about segmentation memory scheme?

Divide program into several logical segments

Use multiple bases and bounds

suffer from internal fragmentation

Need to load entire program into memory

All the given answers are true

48. What is true about paging memory scheme

Page and page frame can be different in size

The smaller the page size is, the smaller the page table is

Larger page transfer as fast as smaller page

Page is in logical memory, page frame is in physical memory

Demand paging is when a program can demand preloading of a page

49. Which memory replacement policy is not really achieved?

Optimal

FIFO

LRU

LFU

All of the answers can be implemented

50. When does a page fault occur

When a process accesses a page that is not valid in page table

When a process access a page that is not in logical memory

When a process access a page that is updated in memory

When a process access a page not updated in memory

All of the answers cause page fault to occur

51. When do you want to use round robin (RR) process scheduling?

In the same level of MLFQ

Users want system to be responsive

Avoid starvation

Round robin are not suitable for all the answers

Round robin can be used in all the situation in the answers